

UD65640

Application Note

TC&TP

AN-K0008-250902

|                                                             |           |
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## 1. Target

Introduce Temperature Sensor & Temperature Compensation & Temperature Protection Register and Flow.

## 2. Register

### TS Control: Temperature Sensors Control Settings (C0h)

| C0h         | TS Control                                                                                                            |                       |    |    |    |          |          |          |          |     |
|-------------|-----------------------------------------------------------------------------------------------------------------------|-----------------------|----|----|----|----------|----------|----------|----------|-----|
|             | W / R                                                                                                                 | B7                    | B6 | B5 | B4 | B3       | B2       | B1       | B0       | Hex |
| Command     | —                                                                                                                     | 1                     | 1  | 0  | 0  | 0        | 0        | 0        | 0        | C0h |
| parameter 1 | W / R                                                                                                                 | 0                     | 0  | 0  | 0  | TS_SE_EN | TS_SW_EN | TS_NE_EN | TS_NW_EN | 00h |
| parameter 2 | W / R                                                                                                                 | TS_PERIOD_COUNT [7:0] |    |    |    |          |          |          |          | 1Eh |
| Description | TS_NW_EN : Enable temperature sensor of NW.<br>1: Enabled ; 0 : Disabled.                                             |                       |    |    |    |          |          |          |          |     |
|             | TS_NE_EN : Enable temperature sensor of NE.<br>1: Enabled ; 0 : Disabled.                                             |                       |    |    |    |          |          |          |          |     |
|             | TS_SW_EN : Enable temperature sensor of SW.<br>1: Enabled ; 0 : Disabled.                                             |                       |    |    |    |          |          |          |          |     |
|             | TS_SE_EN : Enable temperature sensor of SE.<br>1: Enabled ; 0 : Disabled.                                             |                       |    |    |    |          |          |          |          |     |
|             | TS_PERIOD_COUNT [7:0] : Sensing period of temperature sensors.<br>Unit : 2 frame ; Default = 1 second (at FR = 60Hz). |                       |    |    |    |          |          |          |          |     |
|             | 1 = Must fill 1; 0 = Must fill 0; X = Don't care.                                                                     |                       |    |    |    |          |          |          |          |     |
| Restriction |                                                                                                                       |                       |    |    |    |          |          |          |          |     |

### TS Data : Temperature Sensors Data (C1h)

| C1h         | TS Data                                                                                                                                                                                                                                                                                                                                              |    |                   |    |    |    |    |    |    |     |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------------------|----|----|----|----|----|----|-----|
|             | RO                                                                                                                                                                                                                                                                                                                                                   | B7 | B6                | B5 | B4 | B3 | B2 | B1 | B0 | Hex |
| Command     | —                                                                                                                                                                                                                                                                                                                                                    | 1  | 1                 | 0  | 0  | 0  | 0  | 0  | 1  | C1h |
| parameter 1 | RO                                                                                                                                                                                                                                                                                                                                                   | 0  | TS_AVG_CODE [6:0] |    |    |    |    |    |    | 00h |
| parameter 2 |                                                                                                                                                                                                                                                                                                                                                      | 0  | TS_NW_CODE [6:0]  |    |    |    |    |    |    | 00h |
| parameter 3 |                                                                                                                                                                                                                                                                                                                                                      | 0  | TS_NE_CODE [6:0]  |    |    |    |    |    |    | 00h |
| parameter 4 |                                                                                                                                                                                                                                                                                                                                                      | 0  | TS_SW_CODE [6:0]  |    |    |    |    |    |    | 00h |
| parameter 5 |                                                                                                                                                                                                                                                                                                                                                      | 0  | TS_SE_CODE [6:0]  |    |    |    |    |    |    | 00h |
| Description | TS_AVG_CODE [6:0] : All enable TS code’s average value.<br><br>TS_NW_CODE [6:0] : TS_NW_ADC read out code value.<br><br>TS_NE_CODE [6:0] : TS_NE_ADC read out code value.<br><br>TS_SW_CODE [6:0] : TS_SW_ADC read out code value.<br><br>TS_SE_CODE [6:0] : TS_SE_ADC read out code value.<br><br>1 = Must fill 1; 0 = Must fill 0; X = Don’t care. |    |                   |    |    |    |    |    |    |     |
| Restriction |                                                                                                                                                                                                                                                                                                                                                      |    |                   |    |    |    |    |    |    |     |

### TS Trim: Temperature Sensors Trim (C2h)

| C2h         | TS Trim                                                                                                                                                                                                                                                                                                         |      |                       |    |    |    |    |    |    |     |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----------------------|----|----|----|----|----|----|-----|
|             | RO                                                                                                                                                                                                                                                                                                              | B7   | B6                    | B5 | B4 | B3 | B2 | B1 | B0 | Hex |
| Command     | —                                                                                                                                                                                                                                                                                                               | 1    | 1                     | 0  | 0  | 0  | 0  | 1  | 0  | C2h |
| parameter 1 | W/R                                                                                                                                                                                                                                                                                                             | SIGN | TS_NW_TRIM_CODE [6:0] |    |    |    |    |    |    | 00h |
| parameter 2 |                                                                                                                                                                                                                                                                                                                 | SIGN | TS_NE_TRIM_CODE [6:0] |    |    |    |    |    |    | 00h |
| parameter 3 |                                                                                                                                                                                                                                                                                                                 | SIGN | TS_SW_TRIM_CODE [6:0] |    |    |    |    |    |    | 00h |
| parameter 4 |                                                                                                                                                                                                                                                                                                                 | SIGN | TS_SE_TRIM_CODE [6:0] |    |    |    |    |    |    | 00h |
| Description | SIGN: 1= temperature increase, 0= temperature decrease<br>TS_NW_TRIM_CODE [6:0]: temp sensor trim code<br>TS_NE_TRIM_CODE [6:0]: temp sensor trim code<br>TS_SW_TRIM_CODE [6:0]: temp sensor trim code<br>TS_SE_TRIM_CODE [6:0]: temp sensor trim code<br><br>1 = Must fill 1; 0 = Must fill 0; X = Don't care. |      |                       |    |    |    |    |    |    |     |
| Restriction |                                                                                                                                                                                                                                                                                                                 |      |                       |    |    |    |    |    |    |     |

### TC Settings: Temperature Compensation Settings (C3h)

| C3h         | TC Settings                                                                                                                                                                                                                                                                                              |                        |    |    |    |    |         |        |       |     |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|----|----|----|----|---------|--------|-------|-----|
|             | W / R                                                                                                                                                                                                                                                                                                    | B7                     | B6 | B5 | B4 | B3 | B2      | B1     | B0    | Hex |
| Command     | —                                                                                                                                                                                                                                                                                                        | 1                      | 1  | 0  | 0  | 0  | 0       | 1      | 1     | C3h |
| parameter 1 | W / R                                                                                                                                                                                                                                                                                                    | 0                      | 0  | 0  | 0  | 0  | INT_POL | INTOUT | TC_EN | 00h |
| parameter 2 | W / R                                                                                                                                                                                                                                                                                                    | TC_COMPARE_COUNT [7:0] |    |    |    |    |         |        |       | 03h |
| Description | TC_EN : Temperature compensation enable.<br>1: Enabled ; 0 : Disabled.<br><br>INTOUT : HW interrupt.<br>1: Notify DE pin change to output ; 0 : Don't care.<br><br>INT_POL : Rising/falling edge trigger.<br>1: Falling edge ; 0 : Rising edge.<br><br>1 = Must fill 1; 0 = Must fill 0; X = Don't care. |                        |    |    |    |    |         |        |       |     |
| Restriction |                                                                                                                                                                                                                                                                                                          |                        |    |    |    |    |         |        |       |     |

### TC LUT INDEX : TC LUT Number(C4h)

| C4h         | TC Settings                                                                                                     |    |    |    |    |    |                    |    |    |     |
|-------------|-----------------------------------------------------------------------------------------------------------------|----|----|----|----|----|--------------------|----|----|-----|
|             | W / R                                                                                                           | B7 | B6 | B5 | B4 | B3 | B2                 | B1 | B0 | Hex |
| Command     | —                                                                                                               | 1  | 1  | 0  | 0  | 0  | 1                  | 0  | 0  | C4h |
| parameter 1 | W / R                                                                                                           | 0  | 0  | 0  | 0  | 0  | TC_LUT_INDEX [2:0] |    |    | 00h |
| Description | TC_LUT_INDEX [2:0] : Temperature compensation table index.<br>1 = Must fill 1; 0 = Must fill 0; X = Don't care. |    |    |    |    |    |                    |    |    |     |
| Restriction |                                                                                                                 |    |    |    |    |    |                    |    |    |     |

### TC LUT: Temperature Compensation Settings (C5h)

| C5h         | TC Settings                                             |       |                       |    |    |    |                   |    |     |     |
|-------------|---------------------------------------------------------|-------|-----------------------|----|----|----|-------------------|----|-----|-----|
|             | W / R                                                   | B7    | B6                    | B5 | B4 | B3 | B2                | B1 | B0  | Hex |
| Command     | —                                                       | 1     | 1                     | 0  | 0  | 0  | 1                 | 0  | 1   | C5h |
| parameter 1 | W / R                                                   | VAILD | TS_CODE_INDEX [6:0]   |    |    |    |                   |    |     | 00h |
| parameter 2 | W / R                                                   | 0     | ELVSS_SW_PS_NUM [6:0] |    |    |    |                   |    |     | 00h |
| parameter 3 | W / R                                                   | 0     | 0                     | 0  | 0  | 0  | GAMMA_INDEX [2:0] |    | 00h |     |
| Description | TS_CODE_INDEX [6:0] : Temperature value code of TC LUT. |       |                       |    |    |    |                   |    |     |     |
|             | VAILD : TC LUT index valid.<br>1: Used ; 0 : No used.   |       |                       |    |    |    |                   |    |     |     |
|             | ELVSS_SW_PS_NUM [6:0] : Swire pulse number of TC LUT.   |       |                       |    |    |    |                   |    |     |     |
|             | GAMMA_INDEX [2:0] : Gamma curve index of TC LUT.        |       |                       |    |    |    |                   |    |     |     |
|             | 1 = Must fill 1; 0 = Must fill 0; X = Don't care.       |       |                       |    |    |    |                   |    |     |     |
| Restriction |                                                         |       |                       |    |    |    |                   |    |     |     |

### TP Settings: Temperature Protection Settings (C6h)

| C6h         | TC Settings                                                                                                                                                                                                                                                                                                                                                                                                                      |                     |                   |    |    |    |    |           |       |     |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-------------------|----|----|----|----|-----------|-------|-----|
|             | W / R                                                                                                                                                                                                                                                                                                                                                                                                                            | B7                  | B6                | B5 | B4 | B3 | B2 | B1        | B0    | Hex |
| Command     | —                                                                                                                                                                                                                                                                                                                                                                                                                                | 1                   | 1                 | 0  | 0  | 0  | 1  | 1         | 0     | C6h |
| parameter 1 | W / R                                                                                                                                                                                                                                                                                                                                                                                                                            | 0                   | 0                 | 0  | 0  | 0  | 0  | TP_IGNORE | TP_EN | 00h |
| parameter 2 | W / R                                                                                                                                                                                                                                                                                                                                                                                                                            | 0                   | TP_THR_CODE [6:0] |    |    |    |    |           |       | 80h |
| parameter 3 | W / R                                                                                                                                                                                                                                                                                                                                                                                                                            | TP_OVER_COUNT [7:0] |                   |    |    |    |    |           |       | 05h |
| Description | <p>TP_EN: Temperature protection enable.<br/>1: Enabled ; 0: Disabled.</p> <p>TP_IGNORE : Temperature protection operation.<br/>1: No change ; 0: Into Display off.</p> <p>TP_THR_CODE [6:0] : TP threshold value code.</p> <p>TP_OVER_COUNT [7:0] : Period count when temperature over TP_THR_CODE.<br/>Unit: TS_PERIOD_COUNT ; Default = 5 second (at FR = 60Hz).</p> <p>1 = Must fill 1; 0 = Must fill 0; X = Don't care.</p> |                     |                   |    |    |    |    |           |       |     |
| Restriction |                                                                                                                                                                                                                                                                                                                                                                                                                                  |                     |                   |    |    |    |    |           |       |     |

### TEMP FLAG: Temperature Compensation & Protection flag (C7h)

| C7h         | TC Settings                                                                                                                                                                                                                                                |    |    |    |    |    |    |         |         |     |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|----|----|----|---------|---------|-----|
|             | W / R                                                                                                                                                                                                                                                      | B7 | B6 | B5 | B4 | B3 | B2 | B1      | B0      | Hex |
| Command     | —                                                                                                                                                                                                                                                          | 1  | 1  | 0  | 0  | 0  | 1  | 1       | 1       | C7h |
| parameter 1 | W / R                                                                                                                                                                                                                                                      | 0  | 0  | 0  | 0  | 0  | 0  | TP_FLAG | TC_FLAG | 00h |
| Description | <p>TC_FLAG : Temperature compensation flag.<br/>1: TC active ; 0: TC inactive ; write 1 clear</p> <p>TP_FLAG : Temperature protection flag.<br/>1: TP active ; 0: TP inactive ; write 1 clear</p> <p>1 = Must fill 1; 0 = Must fill 0; X = Don't care.</p> |    |    |    |    |    |    |         |         |     |
| Restriction |                                                                                                                                                                                                                                                            |    |    |    |    |    |    |         |         |     |

### 3. Flow

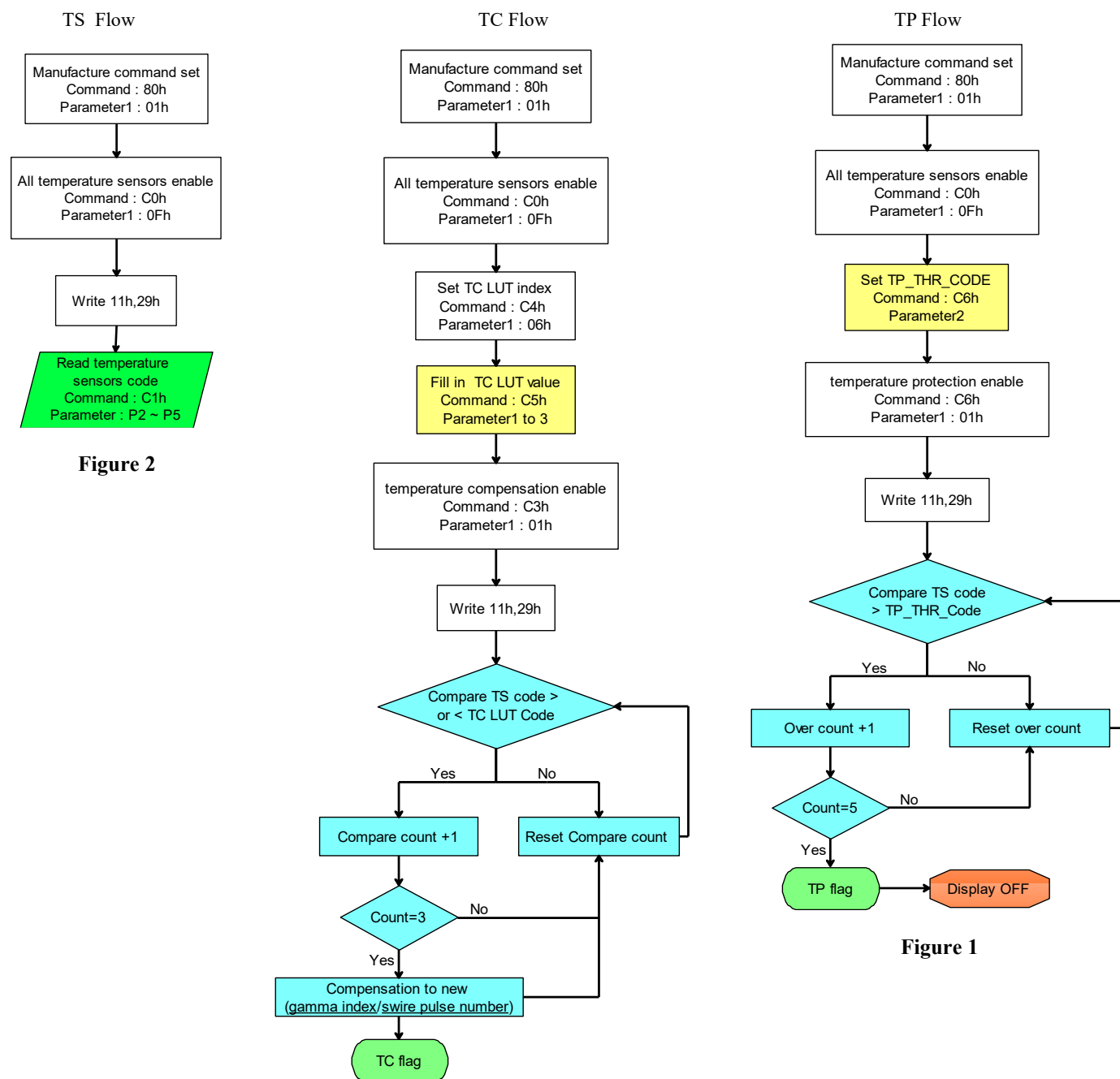


Figure 2

Figure 1

Figure 3



## 4. Example

### 4.1 Enable TS\_NW & TS\_NE

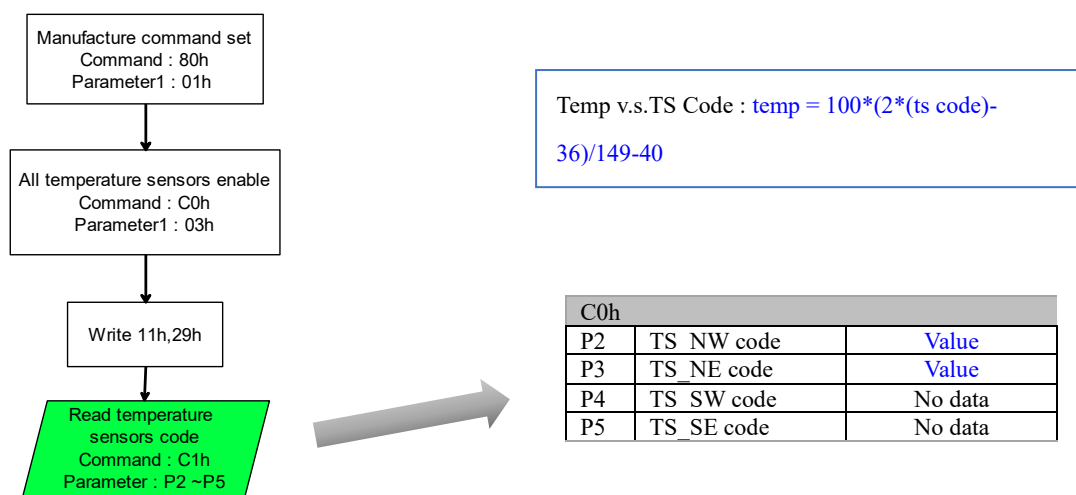


Figure 4

### 4.2 TS Trim Code

| C1h |            |              | Environment |
|-----|------------|--------------|-------------|
| P2  | TS_NW code | 46h (29.8°C) | 29.8°C      |
| P3  | TS_NE code | 34h (5.64°C) | 29.8°C      |

P2 TS\_NE code need trim to environment temperature.

Method:

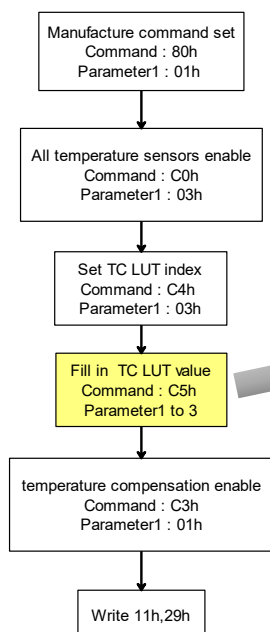
Environment temperature conversion to ts\_code: 46h

Environment temperature ts\_code - TS\_NE code ts\_code = 12h

| C2h |                 |     |
|-----|-----------------|-----|
| P1  | TS_NW TRIM CODE | 00h |
| P2  | TS_NE TRIM CODE | 92h |

Bit7 SIGN 1: Increase, 0: Decreasing.

### 4.3 Enable TC (Enable TS\_NW & TS\_NE) :



**Figure 5**

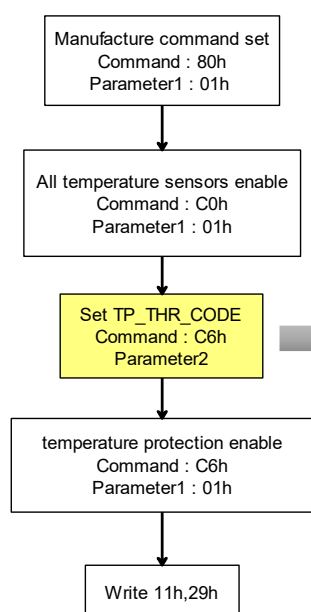
Example LUT :

| LUT Index | Valid | TS code (Code)   | ELVSS (Swire Pulse) | Gamma Index |
|-----------|-------|------------------|---------------------|-------------|
| 0         | 1     | $\geq 0, (<45)$  | 16                  | 0           |
| 1         | 1     | $\geq 45, (<90)$ | 12                  | 1           |
| 2         | 1     | $\geq 90$        | 10                  | 2           |
| 3         | 0     | X                | X                   | X           |
| 4         | 0     | X                | X                   | X           |
| 5         | 0     | X                | X                   | X           |

※ If TC active , verify function by:

1. See panel brightness change or not
2. Measure ELVSS voltage
3. Read TC flag

### 4.4 Enable TP (Enable TS\_NW) :



**Figure 6**

Set TP\_THR\_CODE = 0 code , verify function by :

1. See panel display off or not?
2. Read TP\_flag

## 5. Revision History

| Version No.     | Date      | Page | Description                                      |
|-----------------|-----------|------|--------------------------------------------------|
| AN-K0008-240813 | 2024/8/13 | 9    | Initial Version                                  |
| AN-K0008-250902 | 2025/9/2  | 11   | Add TS_Trim C2h register<br>Add 4.2 TS trim mode |